Structured Requirements Analysis

Part A: Introduction

This task is to model your software system by using structured requirements analysis. There are the following three tasks:

(1) **Data modeling:** Provide the Entity-Relationship (E-R) diagram for your software system.

(2) **Functional modeling:** Provide the Data-Flow Diagram (DFD) for your software system, including at least Level-0, Level-1, and Level-2 diagrams. In addition, use the micro-specification to describe the processes that cannot be decomposed in the given DFD, and use the data dictionary to describe the data-related items such as data flows, data stores, etc.

(3) **Behavioral modeling:** Provide the State Transfer Diagram (SDT) for your software system.

Part B. Writing a Structured Requirements Analysis Specification

(1) Please refer to the template in the Moodle platform. It should be noted that you can adjust the layout of the template by adding some new things or removing some parts from the template.

(2) For all diagrams in your report, use the standard tools to draw them; and it is not acceptable to draw the diagrams by hand.

(3) In the report, the division of labor among group members should be given.

Part C: Submission

Each team is required to submit **one PDF file** into the Moddle platform by the project leader.